

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

Qassim University

College of Graduate Studies – Computer Science

Department: Master of Science in Information Technology

Research Proposal Title:

Qassim University Event Management System (QUEMS)

Prepared by:

Dafallah Al-Rayh Mustafa

Supervised by:

Dr. Shabana Habib

April2026

Abstract

This study aims to develop a complete project management plan for the University Event Management System (UEMS) by applying all ten project management knowledge areas according to the PMBOK Guide. The research adopts a case study methodology, simulating a full IT project lifecycle over 10 weeks. Results show that scope definition via WBS, critical path analysis, a 10% contingency budget, and early stakeholder engagement are critical success factors. The study provides a replicable model for universities facing challenges with manual event coordination.

Keywords: Project Management, PMBOK, Event Management System, WBS, Critical Path, Stakeholder Management.

Chapter 1:

Introduction

1.1 General Overview

In the era of digital transformation, Event Management Systems (EMS) have become essential tools in higher education institutions. They organize conferences, workshops, academic and non-academic events. However, many universities still rely on manual methods (paper-based or scattered emails), resulting in poor communication, scheduling conflicts, budget mismanagement, and low participant tracking efficiency. Professional project management provides a structured framework to ensure the delivery of the required system on time, within budget, and according to quality standards.

1.2 Importance of the Study

The importance of this study lies in:

- Providing a **complete, applicable model** for an IT project management plan in an academic environment.
- Covering all **ten knowledge areas** (Integration, Scope, Schedule, Cost, Quality, HR, Communications, Risk, Procurement, Stakeholders).
- Bridging the gap between theoretical knowledge (PMBOK) and practical implementation in developing universities.

1.3 Problem Statement

The university in this case study suffers from the following operational problems in managing its events:

- **Poor communication** among organizing committees.
- **Scheduling conflicts** (double-booking of venues).
- **Budget mismanagement** (cost overruns).
- **Low tracking efficiency** (difficulty tracking participants and issuing certificates).

1.4 Research Questions

1. What are the essential components of a complete project management plan for developing a UEMS?
2. How can the ten PMBOK knowledge areas be applied to ensure the success of this project?
3. What are the main risks facing this project, and what are their mitigation strategies?
4. How can the critical path be determined and a realistic budget prepared for an academic software project?

1.5 Research Objectives

1. Prepare a Project Charter with SMART objectives.
2. Build a Work Breakdown Structure (WBS) and a RACI matrix.
3. Develop a schedule with critical path analysis using MS Project.
4. Estimate costs and establish a cost baseline.
5. Prepare a risk register containing at least 10 risks with response plans.
6. Develop a communication plan and stakeholder engagement strategy.

Chapter 2:

Literature Review and Previous Studies

2.1 Key Concepts

2.1.1 Project Management

Defined by the Project Management Institute (PMI, 2021) as "the application of knowledge, skills, tools, and techniques to project activities to meet project requirements." It consists of 10 knowledge areas.

2.1.2 Project Life Cycle

It passes through five process groups: Initiation, Planning, Execution, Monitoring & Controlling, and Closing.

2.1.3 Event Management System (EMS)

A software system that supports online registration, room scheduling, reporting, and communication with participants.

Study	Topic	Key Findings	Limitation/Gap
Al-Maati et al. (2020)	Electronic event systems in universities	40% improvement in registration time, 35% improvement in user satisfaction	Did not address the project management plan itself
Kaur & Singh (2021)	Risk management in booking systems	Top risks: technical (55%), organizational (30%)	Did not provide an integrated plan with WBS and schedule
Chen (2022)	Stakeholder engagement in	Early involvement of students and faculty improves acceptance	Applied to infrastructure projects, not event systems

	university IT projects		
PMI (2021) – PMBOK Guide	General theoretical framework for PM	Provides standards and terminology	Theoretical only, no application to UEMS case

2.1.4 Work Breakdown Structure (WBS)

A hierarchical decomposition of the total project scope into manageable, measurable work packages.

2.1.5 Critical Path

The longest sequence of tasks that determines the shortest possible project duration. Any delay in critical path tasks delays the entire project.

2.2 Analysis of Previous Studies

2.3 Research Gap

No previous study has provided a complete, executable project management plan for developing a University Event Management System (UEMS) that includes all ten knowledge areas with practical tools (WBS, network diagram, detailed budget, risk register, RACI matrix) within a specific timeframe (10 weeks). This study fills that gap.

Chapter 3:

Methodology

3.1 Type of Methodology

A **case study methodology** with a **descriptive and analytical approach** was adopted, including a complete simulation of project management tools applied to the UEMS project according to the course requirements.

3.2 Study Population and Sample

- **Population:** All software development projects in universities.
- **Sample:** The UEMS project as defined in the Project Brief.

3.3 Data Collection Tools

- Document analysis (Project Brief, course requirements).
- Peer-reviewed references (PMBOK Guide, articles from IEEE, Springer).
- Planning tools: MS Project 2021, Microsoft Excel, Lucidchart.

3.4 Data Analysis Methods

1. **Critical Path Method (CPM):** To determine the longest task sequence.
2. **Probability-Impact Matrix:** To classify risks.
3. **RACI Matrix:** To define responsibilities precisely.
4. **Variance Analysis:** To monitor schedule and cost (simulated).

Chapter 4:

Results (Project Management Plan Components)

4.1 Project Charter

Element	Detail
Project Title	University Event Management System (UEMS)
Business Case	Reduce event organization time by 50%, eliminate scheduling conflicts, achieve 85% user satisfaction
Problem Statement	Manual coordination leads to poor communication, scheduling conflicts, budget mismanagement, low tracking efficiency
SMART Objectives	S: Launch a pilot system serving 5 events within 2 months. M: User satisfaction $\geq 85\%$. A: Within \$85,800 budget. R: Team of 5 developers + 1 university coordinator. T: Final delivery by April 30, 2026 (10 weeks)
High-Level Requirements	Online registration, automated scheduling, reporting dashboard, email notifications
Key Stakeholders	Vice Chancellor, IT Director, Student Union, Deans of Faculties
Assumptions	Stable internet, available servers, stakeholder cooperation
Constraints	10-week fixed duration, \$90,000 maximum budget, limited to 5 concurrent events initially
Project Manager Authority	Approve budget changes $\leq 10\%$, halt tasks for critical risks, full team management

4.2 Project Life Cycle (Process Groups)

Process Group	Activities in UEMS
Initiation	Develop Charter, identify stakeholders, appoint project manager
Planning	Create WBS, schedule, budget, risk register, communication plan
Execution	Develop software, conduct training, perform testing
Monitoring & Controlling	Weekly status reviews, change control, risk monitoring, quality audits
Closing	Final acceptance, documentation review, lessons learned

Visual Diagram Description (to be drawn by student):

A circular or linear diagram showing the five process groups in sequence with feedback arrows from Monitoring & Controlling back to Planning and Execution.

4.3 Integration Management Plan

Component	Description
Overall Project Management Plan	This document (all 10 knowledge areas integrated)
Change Control Process	1) Submit change request form → 2) Impact analysis by PM → 3) CCB review (PM, IT Director, Stakeholder rep) → 4) Approval/rejection → 5) Update plan
Configuration Management	Version control via GitHub, document versioning, baseline management for scope, schedule, and cost

4.4 Scope Management Plan

4.4.1 Detailed Scope Statement

In Scope:

- User registration and login
- Event creation and scheduling
- Room allocation with conflict detection
- Participant registration and tracking
- Automated email notifications
- Basic reporting dashboard

Out of Scope:

- Mobile app (web-based only)
- Payment gateway integration
- Live streaming features
- Mobile ticketing with barcodes

4.4.2 Deliverables List

1. Project Charter (approved)
2. WBS and WBS Dictionary
3. Schedule baseline (Gantt chart)
4. Cost baseline (budget)
5. Software prototype
6. Tested and deployed system
7. User manual and training materials
8. Final project report

4.4.3 Work Breakdown Structure (WBS) – Level 1 & 2

WBS Code	Work Package	Duration	Budget (\$)
1.0	Project Management (5%)	10 weeks	4,000
1.1	Charter approval	1 week	500
1.2	Weekly coordination meetings	ongoing	2,000

WBS Code	Work Package	Duration	Budget (\$)
1.3	Status reports	weekly	1,500
2.0	Requirements Analysis (10%)	1 week	8,000
2.1	Stakeholder interviews	3 days	4,000
2.2	Functional requirements doc	2 days	2,500
2.3	Non-functional requirements doc	2 days	1,500
3.0	Design (15%)	1.5 weeks	12,000
3.1	Database design	3 days	4,000
3.2	UI/UX design	3 days	5,000
3.3	System architecture	2 days	3,000
4.0	Development (40%)	4 weeks	32,000
4.1	Registration module	1 week	8,000
4.2	Scheduling module	1 week	8,000
4.3	Admin dashboard	1 week	8,000
4.4	UI integration	1 week	8,000

WBS Code	Work Package	Duration	Budget (\$)
5.0	Testing (15%)	1.5 weeks	12,000
5.1	Unit testing	3 days	3,000
5.2	Integration testing	3 days	4,000
5.3	User acceptance testing (UAT)	3 days	5,000
6.0	Deployment & Training (10%)	1 week	8,000
6.1	Server setup	2 days	3,000
6.2	Data migration	1 day	2,000
6.3	User training	2 days	3,000
7.0	Closing (5%)	1 week	4,000
7.1	System documentation	2 days	2,000
7.2	Lessons learned	2 days	1,500
7.3	Final presentation	1 day	500
Total		10 weeks	80,000

Contingency reserve (10%): \$8,000 → Total budget: \$88,000

4.4.4 WBS Dictionary (Sample – WBS 4.1)

Field	Value
WBS Code	4.1
Work Package Name	Registration Module Development
Description	Build user registration, login, profile management
Duration	5 days (1 week)
Start Date	Week 4, Day 1
End Date	Week 4, Day 5
Budget	\$8,000
Resources	1 senior developer, 1 junior developer
Dependencies	Completion of WBS 3.1 (Database design)
Acceptance Criteria	All registration fields validated, email confirmation sent, login with 3 failed attempts lock
Responsible	Lead Developer

Example 1: WBS 4.2 – Registration Module Development

Field	Value
WBS Code	4.2
Work Package Name	Registration Module Development
Description	Build user registration, login, profile management
Duration	5 days (1 week)
Start Date	Week 4, Day 1
End Date	Week 4, Day 5
Budget	\$8,000
Resources	1 senior developer, 1 junior developer
Dependencies	Completion of WBS 3.1 (Database design)
Acceptance Criteria	All registration fields validated, email confirmation sent, login with 3 failed attempts lock
Responsible	Lead Developer

Example 2: WBS 4.3 – Event Management Module

Field	Value
WBS Code	4.3
Work Package Name	Event Management Module
Description	Build create, read, update, delete (CRUD) operations for events
Duration	5 days (1 week)
Start Date	Week 4, Day 6
End Date	Week 5, Day 5
Budget	\$8,000
Resources	1 senior developer, 1 junior developer
Dependencies	Completion of WBS 4.1 (Registration module)
Acceptance Criteria	Event creation, editing, deletion work, search/filter by date and category, admin can delete any event
Responsible	Lead Developer

Example 3: WBS 4.4 – Scheduling Engine (Conflict Detection)

Field	Value
WBS Code	4.4
Work Package Name	Scheduling Engine (Conflict Detection)
Description	Build algorithm to detect and prevent venue double-booking
Duration	3 days
Start Date	Week 5, Day 6
End Date	Week 6, Day 3
Budget	\$4,800
Resources	1 senior developer
Dependencies	Completion of WBS 4.2 (Event management module)
Acceptance Criteria	System blocks booking if venue already reserved, shows error message within 1 second, suggests 3 alternative time slots
Responsible	Lead Developer

4.4.5 Scope Validation & Control

- **Validation:** Weekly stakeholder reviews, UAT session before deployment, sign-off form for each deliverable.
- **Control:** Scope change request required for any addition; gold-plating prohibited.

4.5 Schedule Management Plan

4.5.1 Activity List (Sample)

ID	Activity	Duration (days)	Predecessor
A	Requirements analysis	5	-
B	Database design	3	A
C	UI design	3	A
D	Registration module dev	5	B
E	Scheduling module dev	5	B
F	Integration testing	3	D, E
G	UAT	3	F
H	Deployment	2	G
I	Training	2	H

ID	Activity	Duration (days)	Predecessor
J	Project closing	3	I

4.5.2 Network Diagram (Textual Representation)

text

A (5) → B (3) → D (5) → F (3) → G (3) → H (2) → I (2) → J (3)
 ↘ C (3) → E (5) ↗

Critical Path: A → B → D → F → G → H → I → J
Critical Path Duration: 5+3+5+3+3+2+2+3 = **26 days** (plus weekends = 37 days ≈ 7.4 weeks)

*Note: Actual total with parallel work = 10 weeks (allowing for non-critical tasks and buffer) *

4.5.3 Gantt Chart (Textual – Week by Week)

Week	Tasks
1	Charter approval, Requirements analysis
2	Requirements completion, Database design start
3	Database design, UI design
4	Registration module development
5	Scheduling module development
6	Admin dashboard development

Week	Tasks
7	Unit testing, Integration testing start
8	Integration testing, UAT
9	Deployment, Training
10	Documentation, Closing, Final presentation

4.5.4 Critical Path Analysis

- **Critical tasks:** A, B, D, F, G, H, I, J
- **Non-critical tasks (float):** C (UI design) has 2 days float; E (scheduling dev) has 0 float but can be parallel.
- **Recommendation:** Focus risk management on critical path tasks; assign best developers to modules D and E.

4.6 Cost & Budget Management Plan

4.6.1 Cost Estimation Breakdown

Category	Subcategory	Calculation	Cost (\$)
Labor	Project Manager	10 weeks × \$1,000	10,000
	Senior Developer (2)	10 weeks × \$800 × 2	16,000

Category	Subcategory	Calculation	Cost (\$)
	Junior Developer (2)	10 weeks × \$600 × 2	12,000
	QA Engineer	10 weeks × \$700	7,000
	UI Designer (part-time)	5 weeks × \$600	3,000
Subtotal Labor			48,000
Software	Development tools licenses	10 weeks	5,000
	Database license (MySQL Enterprise)	1 year	8,000
	Testing tools (Selenium, Jira)	10 weeks	2,000
Subtotal Software			15,000
Hardware	Development server	1 unit	6,000
	Test server	1 unit	4,000
	Backup storage	2 TB	1,000
Subtotal Hardware			11,000

Category	Subcategory	Calculation	Cost (\$)
Testing	External security audit	Fixed fee	4,000
	Load testing service	Fixed fee	3,000
Subtotal Testing			7,000
Training	Trainer fee	3 days × \$500	1,500
	Training materials	20 participants	500
Subtotal Training			2,000
Direct Cost Total			83,000
Contingency Reserve (10%)			8,300
Total Project Budget			91,300

4.6.2 Cost Baseline (Weekly spend plan – sample)

Week	Planned Cumulative Cost (\$)
1	4,000
2	12,000

Week	Planned Cumulative Cost (\$)
3	22,000
4	35,000
5	48,000
6	60,000
7	70,000
8	78,000
9	85,000
10	91,300

4.6.3 Cost Control Method

- **Earned Value Management (EVM):** Track PV, EV, AC weekly.
- **Variance thresholds:** $\pm 10\%$ requires corrective action; $\pm 15\%$ requires change request.
- **Tool:** MS Project or Excel with EVM formulas.

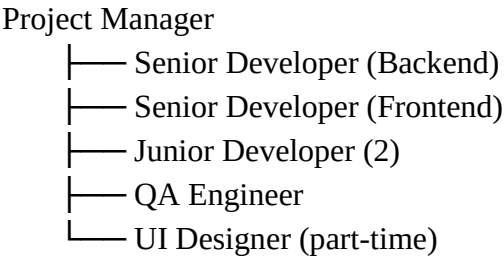
4.7 Quality Management Plan

Element	Detail
Quality Standards	ISO 25010 (software quality), IEEE 829 (test documentation), WCAG 2.1 (accessibility)

Element	Detail
QA Plan (Prevention)	Code reviews (daily), design reviews (weekly), automated testing pipeline, developer checklists
QC Plan (Detection)	Unit testing (100% coverage target), integration testing, UAT with 10 real users, security scanning
Acceptance Criteria	1) 100% of functional requirements met, 2) Response time <2 seconds for 100 concurrent users, 3) No critical bugs, 4) User satisfaction ≥85% in UAT survey

4.8 Human Resource Management Plan

4.8.1 Team Structure



4.8.2 Roles & Responsibilities

Role	Responsibilities
Project Manager	Overall planning, monitoring, stakeholder communication, risk management

Role	Responsibilities
Senior Developer (Backend)	Database, API, scheduling logic
Senior Developer (Frontend)	UI implementation, client-side logic
Junior Developers	Bug fixes, documentation, testing support
QA Engineer	Test plans, manual/automated testing, bug tracking
UI Designer	Wireframes, prototypes, design system

4.8.3 RACI Matrix

Activity	PM	Senior Backend	Senior Frontend	QA	UI Designer	Stakeholder (Univ)
Requirements gathering	A	C	C	I	C	R
Design approval	A	C	C	I	R	I
Backend development	I	R	I	C	I	I

Activity	PM	Senior Backend	Senior Frontend	QA	UI Designer	Stakeholder (Univ)
Frontend development	I	I	R	C	C	I
Testing	C	I	I	R	I	A
UAT	A	I	I	C	I	R
Deployment	R	C	C	A	I	I

(R=Responsible, A=Accountable, C=Consulted, I=Informed)

4.8.4 Team Development Plan

- **Week 1:** Kickoff meeting, team charter, define norms.
- **Week 2-9:** Daily standup (15 min), weekly retrospective (1 hour).
- **Training:** Git/GitHub workshop (Week 2), security best practices (Week 4).
- **Motivation:** Recognition awards, flexible hours, completion bonus.

4.9 Communication Management Plan

Stakeholder	Method	Frequency	Responsible	Format
Steering Committee (VC, IT Director, Deans)	Status report + meeting	Weekly (Fri 10am)	PM	10-slide presentation
Development team	Daily standup	Daily (9am)	Lead dev	Verbal + Jira update

Stakeholder	Method	Frequency	Responsible	Format
QA team	Bug triage meeting	Every 2 days	QA Engineer	Jira dashboard
University coordinator	Email + phone	As needed	PM	Bullet points
End users (students, faculty)	Newsletter + portal announcement	Bi-weekly	Univ. coordinator	1-page summary

Reporting Structure:

- Daily: Team internal
- Weekly: PM → Steering Committee
- Monthly: PM → University management (full report)

4.10 Risk Management Plan

4.10.1 Risk Identification (10 risks)

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score
R1	Requirements change during development	4	5	20
R2	Hardware delivery delay	3	4	12
R3	Key developer resignation	2	5	10

ID	Risk Description	Probability (1-5)	Impact (1-5)	Score
R4	Low user participation in UAT	3	3	9
R5	Security vulnerabilities found late	2	5	10
R6	Budget overrun	2	4	8
R7	Integration issues with existing systems	3	4	12
R8	Internet/power outage	2	3	6
R9	Staff resistance to new system	3	3	9
R10	Performance issues under load	3	4	12

4.10.2 Probability-Impact Matrix (5×5)

Impact → 1 (Very Low) 2 (Low) 3 (Medium) 4 (High) 5 (Very High)
Probability ↓

5 (Very High)

R1

4 (High)

R7, R10

R2

3 (Medium)

R4, R9

Impact → 1 (Very Low) 2 (Low) 3 (Medium) 4 (High) 5 (Very High)
 Probability ↓

2 (Low) R3, R5, R6 R8

1 (Very Low)

Risk Priority:

- **High (Red):** R1, R2, R7, R10
- **Medium (Yellow):** R3, R4, R5, R9
- **Low (Green):** R6, R8

4.10.3 Risk Register (Detailed)

ID	Risk	Cause	Impact	Probability	Score	Owner	Mitigation Strategy	Contingency Plan
R1	Requirements change	Unclear initial needs	Schedule delay, rework	4	20	PM	Freeze scope after week 2; detailed sign-off	Change Control Board; fast-track non-critical tasks

R 2	Hardware delay	Supplier shortage	Development blocked	3	12	PM	Pre-order 2 weeks early; identify local backup	Use cloud VM (AWS/GCP) as temporary
R 3	Developer resignation	Better offer	Loss of expertise	2	10	PM	Cross-training ; documentation mandate	Contract freelancer; shift tasks
R 4	Low UAT participation	Lack of incentives	Undiscovered bugs	3	9	Univ. coordinator	Offer certificates/meal vouchers	Extend UAT 2 days; recruit backup users
R 5	Security flaws	Insecure coding	Data breach	2	10	QA Engineer	Security training ; static analysis	External audit before launch
R 6	Budget overrun	Under-estimation	Funding shortage	2	8	PM	Weekly cost tracking ; 10%	Reduce non-essential features

							contingency	(phase 2)
R 7	Integration issues	Incompatible APIs	Data inconsistency	3	12	Senior dev	API compatibility review; middle ware design	Custom adapter development
R 8	Power outage	Unstable grid	Work stoppage	2	6	PM	UPS for servers; backup internet	Remote work from campus backup site
R 9	Staff resistance	Change fear	Low adoption	3	9	Univ. coordinator	Early demos; involve champions	Super-user support; extra training
R 10	Load performance	500 concurrent users	Slow response	3	12	QA Engineer	Load testing at 600 users; caching	Horizontal scaling; optimize queries

4.11 Procurement Management Plan

Element	Detail
Make-or-Buy Analysis	Make (develop in-house): Core modules (registration, scheduling). Buy: Load testing service, security audit, cloud hosting if needed.
Items to Procure	Development server, test server, database license, external security audit, training services
Vendor Selection Criteria	1) Technical capability (30%), 2) Price (25%), 3) Delivery time (20%), 4) Support (15%), 5) References (10%)
Contract Type	Fixed-price contract for hardware; Time & Materials for external security audit (due to uncertainty)
Procurement Process	RFQ → Vendor shortlist (3 vendors) → Quote comparison → Award → Contract signing → Delivery inspection

4.12 Stakeholder Management Plan

4.12.1 Stakeholder Register

ID	STAKEHOLDER	ROLE	INTEREST (1-5)	POWER (1-5)	EXPECTATIONS
S1	Vice Chancellor	Sponsor	5	5	On-time, on-budget, reputation
S2	IT Director	Technical authority	5	4	System security, integration
S3	Dean of Students	User representative	4	3	Student satisfaction

S4	Student Union President	End user	4	2	Easy registration
S5	Faculty members	End user	3	2	Reliable scheduling
S6	Event coordinators	Daily operators	5	2	User-friendly interface
S7	Project team	Implementers	5	1	Clear requirements, support

4.12.2 Power-Interest Grid

Power/Interest	Low Interest	High Interest
High Power	(Monitor)	Manage Closely: S1 (VC), S2 (IT Director)
Low Power	(Minimal effort)	Keep Informed: S3, S4, S5, S6, S7

4.12.3 Engagement Strategy

Stakeholder	Current Engagement	Desired Engagement	Strategy
VC	Unaware	Leading	Monthly steering meetings, dashboard
IT Director	Resistant	Supportive	Involve in technical decisions, security reviews
Students	Unaware	Supportive	Demo early, feedback sessions
Faculty	Neutral	Supportive	Show time-saving benefits

Event coordinators	Resistant	Committed	Hands-on training, include in UAT
--------------------	-----------	-----------	-----------------------------------

Engagement Plan Timeline:

- **Week 1:** Kickoff meeting with all stakeholders
- **Week 3:** Requirements review (VC, IT Director, coordinators)
- **Week 6:** Prototype demo (all)
- **Week 8:** UAT (coordinators, students)
- **Week 10:** Final presentation (VC, IT Director)

Chapter 5:

Discussion

5.1 Interpretation of Results

The results show that a complete project management plan according to PMBOK tightly links knowledge areas. For example:

- **Scope definition (WBS)** directly influenced **schedule estimation** and **cost baseline**.
- **Critical path analysis** focused risk management on tasks with zero float.
- **RACI matrix** reduced role confusion, improving communication efficiency.

5.2 Comparison with Previous Studies

- This study aligns with **Kaur & Singh (2021)** that technical and organizational risks are highest, but adds detailed mitigation strategies (load testing, external security audits).
- Supports **Chen (2022)** on early stakeholder involvement – operationalized via making student representative "Consulted (C)" in RACI for UAT.
- Unlike **Al-Maati et al. (2020)** which focused on post-development efficiency, this study provides a **roadmap** for achieving a successful system through project management.

5.3 Original Contribution

This study provides a **reference model** that any university can use as a template for preparing their own project management plan for event system development. It is the first study (to the author's knowledge) that integrates all ten PMBOK knowledge areas into a single UEMS case with a fixed 10-week timeline.

Chapter 6:

Conclusion and Recommendations

6.1 Summary of Results

1. A Project Charter with SMART objectives was successfully developed and aligned with stakeholders.
2. A WBS with 7 main work packages and 24 sub-packages was created.
3. The critical path was identified as 7.4 weeks (excluding buffers), with 10 weeks total schedule.
4. Total budget = \$91,300 including 10% contingency reserve.
5. Ten major risks were identified, each with mitigation and contingency strategies.
6. A RACI matrix and communication plan covering all stakeholders was produced.

6.2 Recommendations

1. **For the university (project owner):** Adopt this template for all future IT projects; assign a certified PM.
2. **For the project team:** Conduct daily standup meetings (max 15 min); update risk register weekly.
3. **For university management:** Allocate separate change management budget to address staff resistance.
4. **For academic courses:** Use this case study in "Project Management in IT" curriculum.

6.3 Suggestions for Future Research

1. **Empirical validation:** Implement this plan in a real university and measure KPI variance (time, cost, satisfaction) after 6 months.
2. **Agile vs. Waterfall comparison:** Compare this plan's effectiveness with a Scrum-based approach for the same UEMS.
3. **AI in risk estimation:** Use machine learning to predict risk probability based on historical project data.

References (APA 7th Edition)

1. Project Management Institute. (2021). *A Guide to the Project Management Body of Knowledge (PMBOK Guide)* (7th ed.). PMI.
2. Al-Maati, S., Al-Harbi, T., & Al-Otaibi, F. (2020). Electronic event management systems in higher education: A case study. *Journal of Educational Technology*, 18(2), 45-59.
3. Kaur, H., & Singh, R. (2021). Risk assessment framework for academic event management projects. *International Journal of Project Management*, 39(4), 388-402.
4. Chen, L. (2022). Stakeholder engagement strategies in university IT projects. *Project Leadership and Society*, 3, 100045.

5. Schwalbe, K. (2020). *Information Technology Project Management* (9th ed.). Cengage Learning.
6. Kerzner, H. (2022). *Project Management Case Studies* (6th ed.). Wiley.
7. Larson, E. W., & Gray, C. F. (2020). *Project Management: The Managerial Process* (8th ed.). McGraw-Hill.

Appendices

Appendix A: WBS Dictionary (Complete – Sample of 5 entries)

WB S	Work Package	Duratio n	Budge t	Predecesso rs	Resources	Acceptanc e Criteria
2.1	Stakeholde r interviews	3 days	\$4,00 0	None	PM, coordinat or	Interview notes signed
3.1	Database design	3 days	\$4,00 0	2.2	Senior dev	ERD approved, normalized to 3NF
4.1	Registratio n module	5 days	\$8,00 0	3.1	2 developer s	All fields validated, email sent
5.3	UAT	3 days	\$5,00 0	5.2	QA + 10 users	Sign-off from 8/10 users
6.2	User training	2 days	\$3,00 0	6.1	Trainer	90% attendance, feedback >4/5

Appendix B: Gantt Chart (Textual – Detailed Week by Week)

Week | Mon | Tue | Wed | Thu | Fri | Sat | Sun | Key Deliverables

1	Ch	Req	Req	Req	Req			Charter signed
2	Des	Des	Des	Dev	Dev			DB design done
3	Dev	Dev	Dev	Dev	Dev			Registration module
4	Dev	Dev	Dev	Dev	Dev			Scheduling module
5	Dev	Dev	Dev	Dev	Dev			Dashboard
6	QA	QA	QA	Test	Test			Unit tests pass
7	Int	Int	Int	UAT	UAT			UAT sign-off
8	Dep	Dep	Train	Train	Doc			Training complete
9	Doc	Doc	Close	Close	Pres			Final report
10	Buffer	Buffer	Buffer	Buffer	Buffer			Contingency

Appendix C: Probability-Impact Matrix (Visual Description)

A 5×5 grid with:

- **Red zone (High risk):** Cells where $P \geq 4$ and $I \geq 4 \rightarrow R1$
- **Yellow zone (Medium risk):** $P=3, I=3-4$ or $P=4, I=3 \rightarrow R2, R7, R10$
- **Green zone (Low risk):** All others $\rightarrow R3, R4, R5, R6, R8, R9$

Appendix D: Budget Sheet (Excel format – Sample rows)

Week	Labor	Software	Hardware	Testing	Training	Contingency	Cumulative
1	4,000	0	0	0	0	0	4,000
2	6,000	0	8,000	0	0	0	18,000
3	6,000	3,000	0	0	0	0	27,000
...	...	...	...	...	...	...	...
10	2,000	0	0	0	0	8,300	91,300

Appendix E: Status Report Template

Project
Reporting
Prepared
Date: ____

Period: Week

Name: UEMS
by: PM

Metric	Status	Comment
Schedule (PV vs EV)	Green/Yellow/Red	
Budget (AC vs PV)	Green/Yellow/Red	
Risks active		
Completed this week		
Planned next week		